

Andrew Smithies AIA C.Act MMathStat

Actuary turned Data Scientist

+44 (0) 775-438-1792 | a.smithies@me.com | [linkedin.com/in/andrew-g-smithies](https://www.linkedin.com/in/andrew-g-smithies) | asmithies-dsi.github.io/

MOTIVATION

- Passionate about solving commercial problems using Data Science & Analytics.
- Qualified actuary with experience in General insurance reserving, capital and pricing.
- Natural communicator with excellent grammar, punctuation and syntax.
- I am constantly **learning**, and always looking to **improve**.

SKILLS & TOOLS

Programming: SQL, Python (Base, Pandas, Numpy, Matplotlib, Scikit-Learn, TensorFlow, PySpark), VBA

Tools: Excel, Tableau, Github, Access, DAX, ResQ, Arius, ASTRA, SolvencyTool, AWS (S3, Lambda, IAM, EC2, SageMaker, RDS, DynamoDB)

Statistics: Hypothesis Testing, A/B Testing, Central Limit Theorem, Distributions

Machine Learning: Linear Regression, Logistic Regression, Decision Trees, Random Forest, KNN, k-means, PCA, Association Rule Learning, Causal Impact Analysis, Neural Networks, LLMs, Generative AI, Agentic AI

WORK EXPERIENCE

Reserving & Capital Actuary, SAGA/Ageas (London - remote) Sep 2024 - Jul 2026

- Led production of papers for quarterly Reserving Committee and monthly Large Loss Committee; including actuarial best estimates, IFRS17 LIC, **Stochastic Reserve Variability** Governance, **Reserve Risk & Margin (Scenario testing)**, and comparisons to independent actuaries' results.
- Led deep-dive reviews into development patterns for TPD, TPI (Small) and Large Loss perils, leading to **reduced manual workload and improved accuracy** of reserve estimates.

Actuarial Manager, PKF Littlejohn LLP (Canary Wharf, London - hybrid) Sep 2022 - Sep 2024

- **Project-managed independent reserve reviews** for a diverse range of insurance clients.
- To improve the firm's audit offering for clients with occupational disease exposures, I developed a **frequency-severity projection** tool in **Excel** for occupational disease exposures, enabling the firm to make its own independent estimates rather than just commenting on clients' methodology & assumptions.
- Designed **sensitivity tests** to check that a specialist reinsurance client's in-house built **SQL**-based reserving application was working as expected.
- Reviewed the **actuarial models, cashflow projections** and key actuarial inputs implemented by a client to prepare their IFRS 17 Day 1 balance sheet position.
- **Actuarial expert** supporting internal audits of both pricing and reserving processes for a managing agent.

Reserving Actuary, ZEGO (Shoreditch, London - remote) Sep 2021 - Aug 2022

- Implemented prototype tools to bring quarterly reserving processes in-house, and worked with the BI team to scale up **warehousing of historical data**.
- Collaborated with the Claims team in **building analytics & ML** to monitor third party administrators and spot **trends & outliers** in their bordereaux and claims handling processes.
- Developed quarterly **dashboards** on claims inflation and underwriting year performance.

Business Transformation Actuary, AXA UK plc (City of London, hybrid) Nov 2018 - Sep 2021

- Designed and built prototype **SQL/VBA cashflow projection & discounting** model for IFRS17 & Solvency II reporting, then **led team of Data Engineers** to scale it up in the cloud using **DataBricks, Python & Azure**.
- Assisted with transformation of reserving processes via a new **centralised data warehouse**

Circulation Analyst, KHL Group LLP (Wadhurst, E. Sussex) Jun 2018 - Nov 2018

- Built target models for **readership segmentation** based on GDP and construction industry indicators for reporting actual performance versus targets.
- Implemented changes to **improve efficiency and quality of forecasting** processes and data management.

Operational Planning Analyst, Pearson plc (Westminster, London) Mar 2018 - May 2018

- Developed **KPI dashboard solutions** in **Tableau** to track compliance with internal security standards and improving product release cycle efficiency.
- **Time series modelling of demand vs capacity** for service teams, to help them with prioritising resources.

WORK EXPERIENCE (CONTINUED)

Senior Digital Analytics Analyst, Reach plc (Canary Wharf, London) *Feb 2017 - Mar 2018*

- Developed new KPI scorecard for product teams in **Tableau** with performance benchmarking metrics.
- Ran training sessions for in-house analytics tool users.

Principal Actuarial Analyst, Direct Line Group plc (Bromley, S. London) *Jun 2016 - Jul 2016*

Senior Actuarial Analyst, Sterling Insurance (Kings Hill, Kent) *Jan 2012 - Jun 2016*

- Produced **monthly analytics & reports** for Household & Commercial Product/Pricing Committee, led a project to migrate them to a new database, reviewed pricing models & change proposals for approval.
- Developed **Excel linear regression models** of loss ratio, claim frequency and Catastrophe loadings for independent actuarial challenge to FP&A budget assumptions.

Actuarial Analyst, Assicurazioni Generali SpA - UK Branch (Tower Hamlets, London) *Mar 2011 - Jan 2012*

Actuarial Analyst, RSA Insurance Group plc (Horsham, W. Sussex) *May 2008 - Mar 2011*

- Redesigned **SQL** scripts to collect data off the in-house claims handling systems; then reconciled and built the team's new improved **Access database**.
- Updated postcode rating tables in **SAS**.

Analyst, Certus Ltd (Royal Tunbridge Wells, Kent) *Aug 2004 - Jan 2008*

- Maintained a **Monte Carlo simulation model** of Engineering project losses for a client and their brokers.
- Led an analysis of Property and Business Interruption claims for a mutual Energy P&I Club, used to derive **annual cost & frequency distributions** for both the club's member pool, and its parent group.

PROJECTS

Quantifying Impact of "Delivery Club"

- Used **causal impact analysis** on grocery transaction data to analyse & understand the sales uplift for customers who joined a new service offering free deliveries for an annual fee following a mailer campaign.

Assessing Campaign Performance Using Chi-Square Test For Independence

- Used **A/B testing** to assess the performance of two types of mailers that were sent out to promote the Delivery Club described above. This shows whether the higher signup rate from the more expensive mailer is statistically significant, allowing the client to make more informed decisions and optimise campaign ROI.

"You Are What You Eat" Customer Segmentation

- Used **k-means clustering** on grocery transaction data to split out customers into distinct "shopper types", and to more accurately target customers with relevant content & promotions.

COURSES, CERTIFICATES AND QUALIFICATIONS

Data Science & AI Professional Certification

Mar 2026 - Sep 2026

Data Science Infinity, UK

- **Actionable Learnings:** Extracting & manipulating data using SQL. Application of statistical concepts such as hypothesis tests for measuring the effect of A/B Tests. Github for version control and collaboration. Python for data analysis, manipulation & visualisation. Data preparation steps for ML including missing values, categorical variable encoding, outliers, feature scaling, feature selection & model validation. Machine Learning algorithms for regression, classification, clustering, association rule learning, and causal impact analysis for measuring the impact of an event over time. Machine Learning pipelines to streamline the ML pre-processing & modelling phase. Deployment of an ML pipeline onto a live website using Streamlit. Tableau for powerful Data Visualisations. Turning business problems into Data Science solutions.

AIA C.Act (Chartered Associate Status)

2017

Institute & Faculty of Actuaries, UK

EDUCATION

MMathStat (Class 2:1 Honours)

2000 - 2004

University of Warwick, Coventry

- Undergraduate Masters degree jointly majoring in Mathematics and Statistics.

INTERESTS

Appearing on TV/Radio quiz shows (Weakest Link, Fifteen to One, Brainteaser, Memory Bank, Eggheads, Wordplay, Mastermind, Only Connect, The Chase, Countdown, Brain of Britain and The Finish Line to date). I compete in three quiz leagues and a tenpin bowling league, plus I'm a keen photographer & darts player. I recently rediscovered my childhood passion for LEGO, and I'm currently using it to build a model train layout.